Lecture 41

# Wrapping Things Up

CS61B, Fall 2025 @ UC Berkeley

# 61B in 12 Minutes or Less

Lecture 41, CS61B, Fall 2025



**61B in 12 Minutes or Less**

Your Future

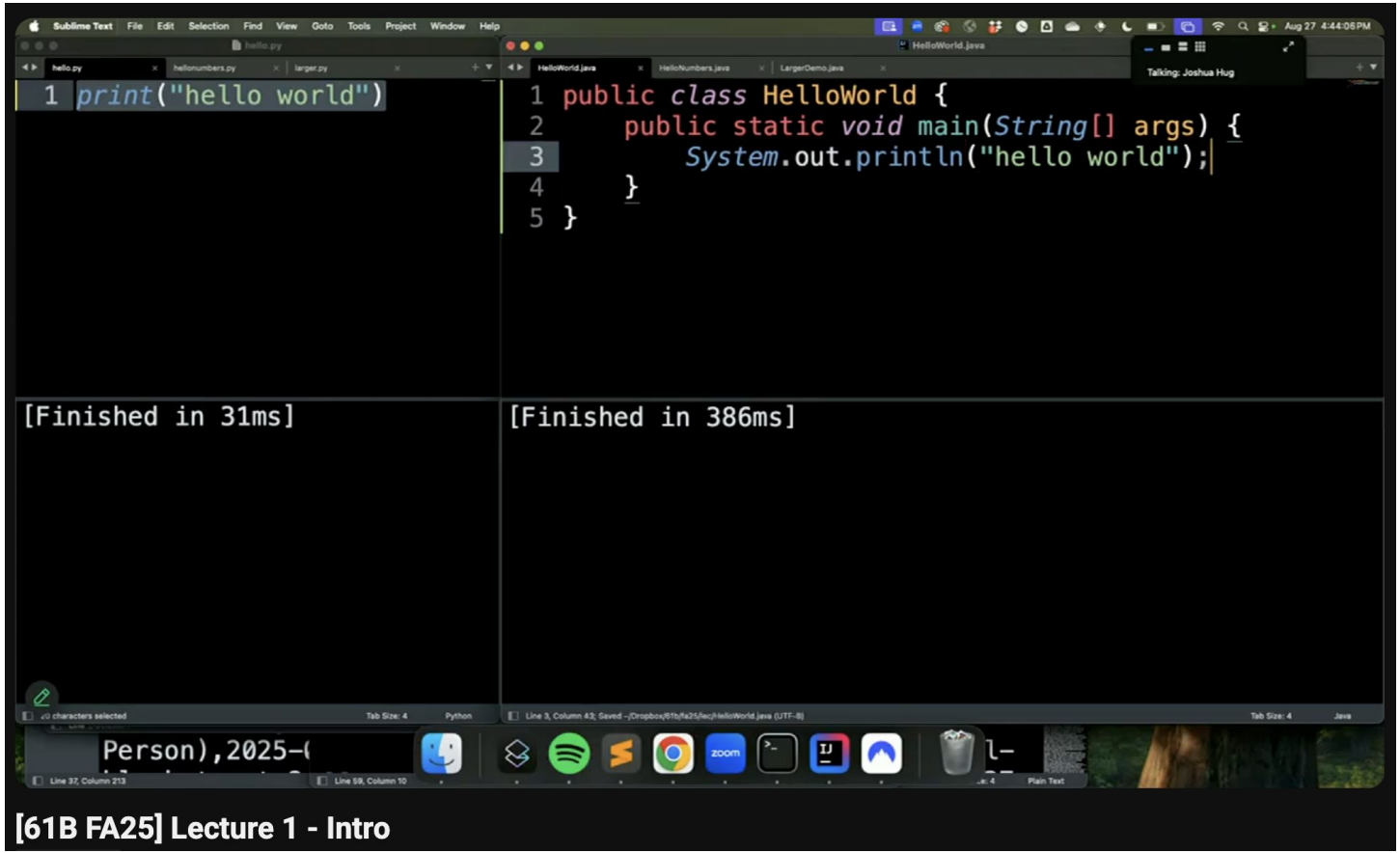
Course Reflections

AMA

So Long!

# Where We Started

Just 14 weeks ago:



# What We've Learned about Programming Languages

- Object oriented programming:
  - Interface Inheritance.
  - Implementation inheritance.
- Generic Programming, e.g. ArrayList<Integer>, etc.
- The model of memory as boxes containing bits.
- Bit representation of positive integers (hashing lecture).
- Java.
- Some standard programming idioms/patterns:
  - Objects as function containers (e.g. Comparators).
  - Default method specification in interfaces.
  - Iterators.

**Example: Programmer only needs to know List API, doesn't have to know that ArrayList secretly does array resizing.**

**Example: Array is a sequence of boxes. An array variable is a box containing address of sequences of boxes.**

# Important Data Structures in Java

Important data structure interfaces:

- java.util.List
- java.util.Set
- java.util.Map
- Our own Collections (e.g. Map61B, Deque61B)

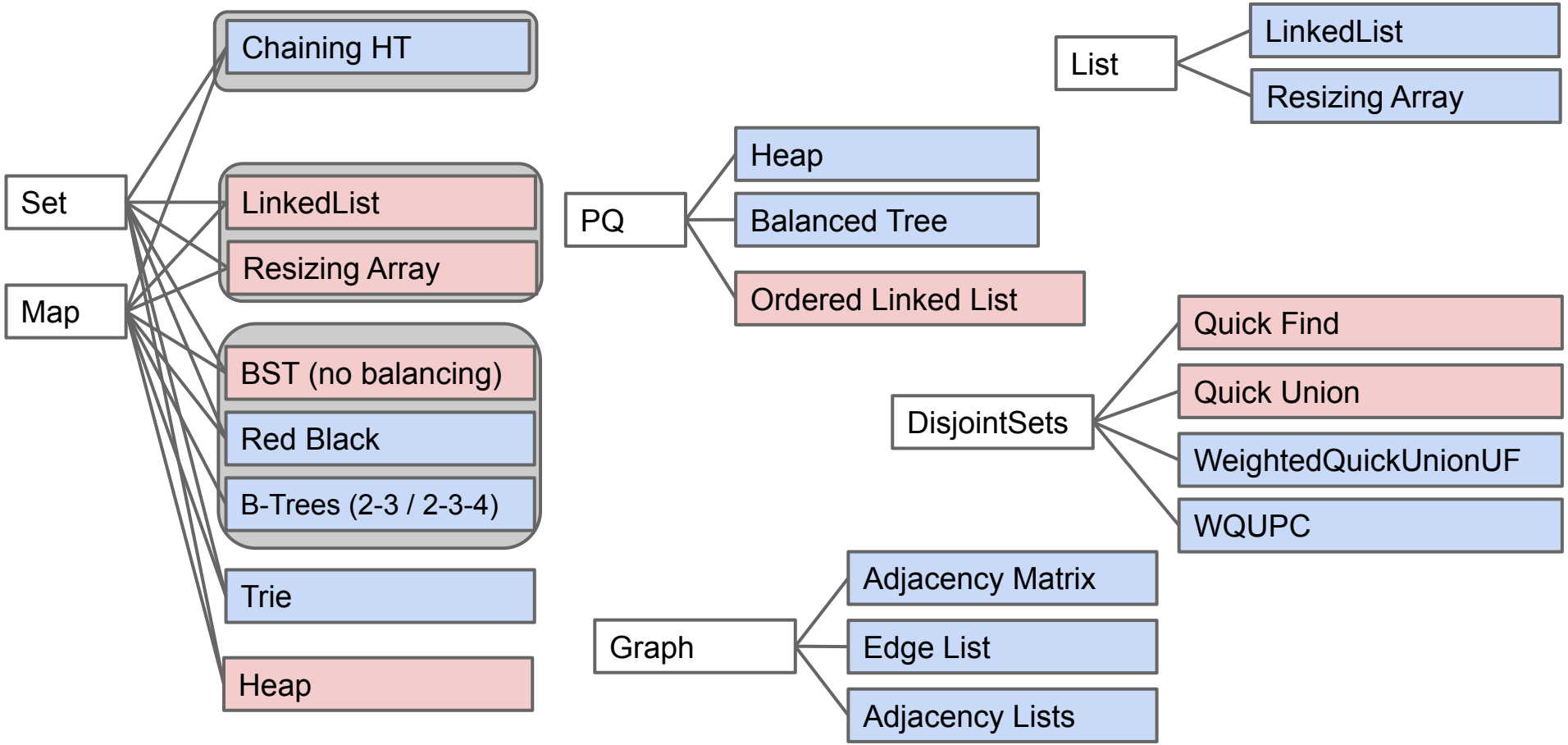
Concrete implementations of these abstract implementations:

- Examples: ArrayDeque61B implements Deque61B.

# Mathematical Analysis of Data Structure/Algorithm Performance

- Asymptotic analysis.
- $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$.
- Worst case vs. average case vs. best case.
  - Exemplar of usefulness of average case: Quicksort
- Determining the runtime of code through inspection (often requires deep thought).
- Multiplicative resizing is much better than additive resizing.
  - Multiplicative resizing results in $\Theta(1)$ add runtime for ArrayLists.
  - Additive resizing results in $\Theta(N)$ add runtime for ArrayLists.

# Some Examples of Implementations for ADTs



# Arrays vs. Linked Data Structures

Array-Based Data Structures:

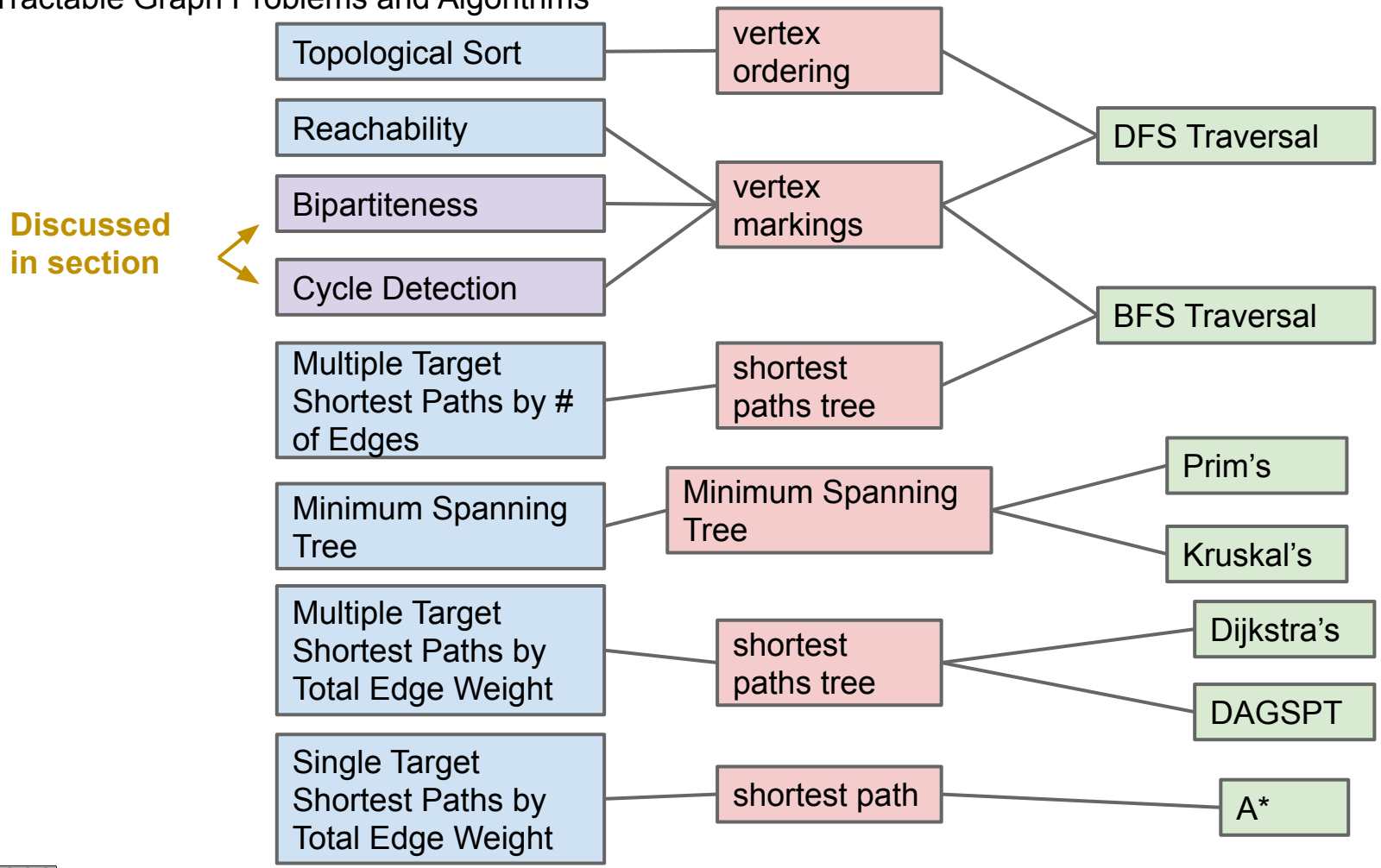
- ArrayLists and ArrayDeque
- HashSets, HashMaps, MyHashMap: Arrays of ‘buckets’
- ArrayHeap (tree represented as an array)

Linked Data Structures

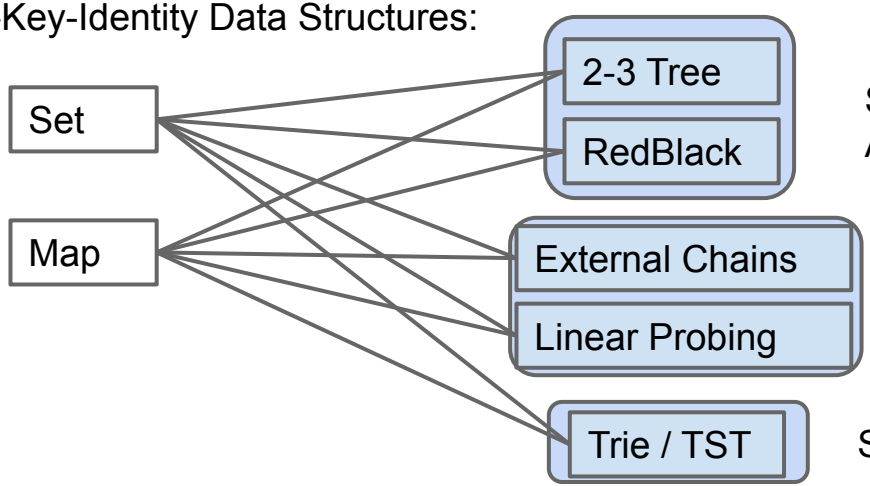
- Linked Lists
  - LinkedList, IntList, LinkedListDeque, SLList, DLList
- Trees: Hierarchical generalization of a linked list. Aim for bushiness.
  - TreeSet, TreeMap, BSTMap, Tries (trie links often stored as arrays)
- Graphs: Generalization of a tree (including many algorithms).

Tradeoffs of array-based vs. linked data structures.

# Tractable Graph Problems and Algorithms



Search-By-Key-Identity Data Structures:

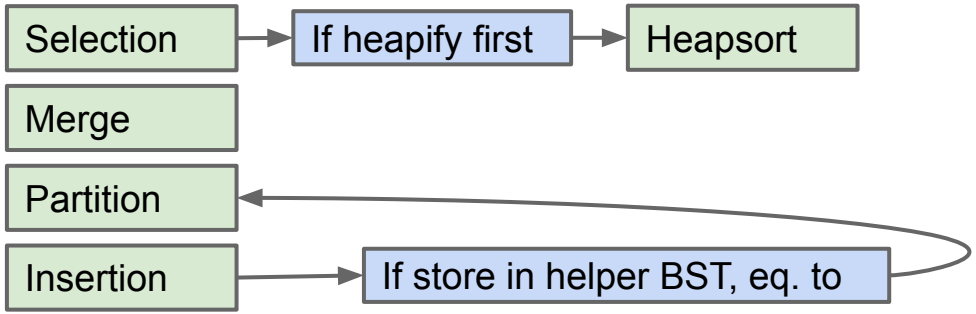



Searches using compareTo()
Analogous to **Comparison-Based**

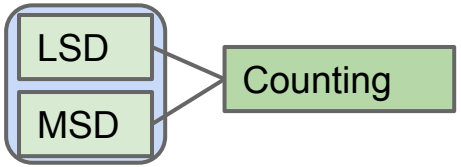
Searches using hashCode() and equals()
Roughly Analogous to **Integer Sorting**

Searches by digit. Analogous to **Radix Sorting**

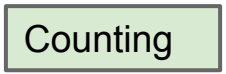
**Comparison Based** Sorting Algorithms: Ω(N log N) worst case




**Radix Sorting** Algorithms: Θ(NW) worst case:
(require a sorting subroutine)




Small-Alphabet **(Integer) Sorting Algorithms**: Θ(N) worst case



# The Practice of Programming

- Java syntax and idioms.
- Unit testing (and its more extreme form: Test-driven development).
- Asking the internet and LLMs for help.
- Debugging:
  - Identify the simplest case affected by the bug.
  - Hunt it down, giving it no place to hide.
  - With the right methodology, can find bugs even when finding bug through manual code inspection is impossible.
- Tactical vs. Strategic Programming.
  - Avoid information leakage. Minimize dependencies. Minimize obscurity.
- Real tools: IntelliJ, git, Truth, command line.
- Data structure selection (and API Design)
  - Drive the performance and implementation of your entire program.

# Your Future

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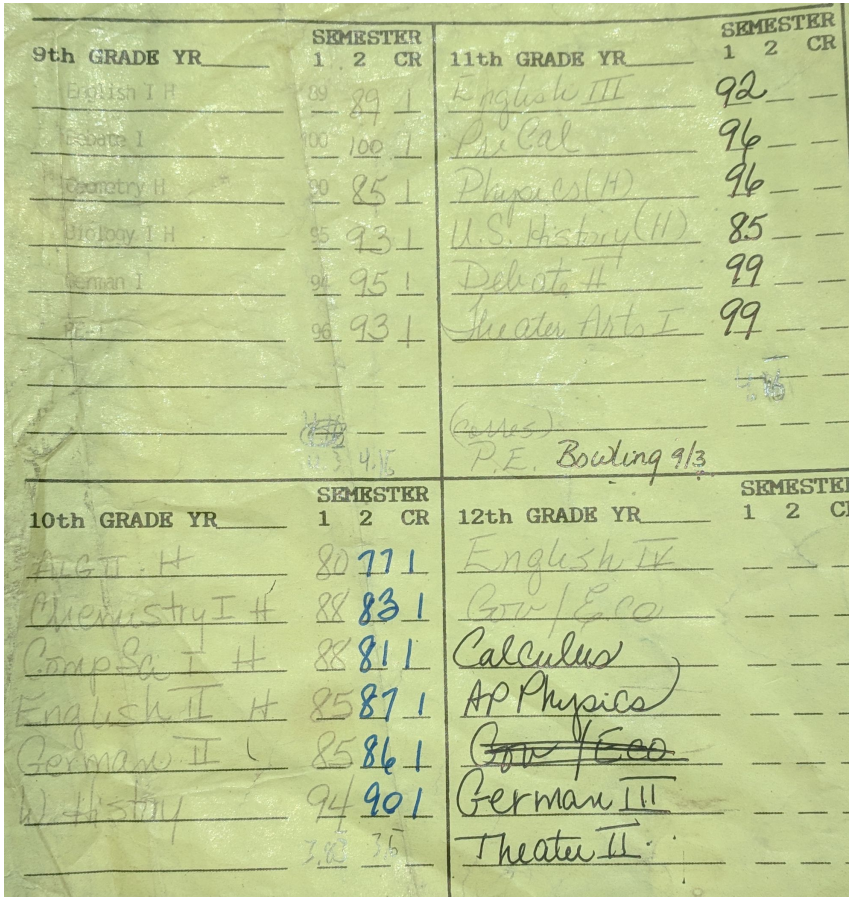
# Thoughts on the Future

Two interesting questions:

- What can you do?
- What should you do?

# Some Josh Hug History



| 9th GRADE YR | SEMESTER 1 | 2 | CR |
|---|---|---|---|
| English I H | 89 | 89 | 1 |
| Geometry I | 100 | 100 | 1 |
| Geometry H | 90 | 85 | 1 |
| Biology I H | 95 | 93 | 1 |
| German I | 94 | 95 | 1 |
| P.E. | 96 | 93 | 1 |
| | | | |
| | | | |
| | | | |

| 11th GRADE YR | SEMESTER 1 | 2 | CR |
|---|---|---|---|
| English III | 92 | | |
| Pre Cal | 96 | | |
| Physics (H) | 96 | | |
| U.S. History (H) | 85 | | |
| Debate II | 99 | | |
| Theater Arts I | 99 | | |
| | | | |
| (Gym) | | | |
| P.E. Bowling 9/3 | | | |

| 10th GRADE YR | SEMESTER 1 | 2 | CR |
|---|---|---|---|
| AlgII H | 80 | 77 | 1 |
| Chemistry I H | 88 | 83 | 1 |
| Comp Sci I H | 88 | 81 | 1 |
| English II H | 85 | 87 | 1 |
| German II | 85 | 86 | 1 |
| World History | 94 | 90 | 1 |
| | | | |

| 12th GRADE YR | SEMESTER 1 | 2 | CR |
|---|---|---|---|
| English IV | | | |
| Gov/Eco | | | |
| Calculus | | | |
| AP Physics | | | |
| ~~Gov/Eco~~ | | | |
| German III | | | |
| Theater II | | | |

# Your Future Capability

Quoting Justin Yokota: “I would argue that the most important thing your grade tells you is how hard your next semester will be.

Most of the time, you don't truly master a concept until you're forced to use it to solve something new. A lot of the concepts we've covered in the last half of the course will only solidify when you start taking 61C, 170, or any other "next step" class.

The grade you get in this course is my message to you saying how difficult I predict it will be to develop that mastery in your next class.”

My strong belief: With sufficient high quality deliberate practice, you can all get **very good** at any topic you study here at Berkeley.

# Your Future Career

You are a rare commodity.

Sources: Link, Link

Revenue

| | |
|---|---|
| Meta | $164.5 billion |
| Alphabet (Google) | $350.0 billion |
| Amazon | $638.0 billion |
| Microsoft | $245.0 billion |
| Apple | $391.0 billion |

Revenue per Employee

| | |
|---|---|
| Meta | $2,221,000 |
| Alphabet (Google) | $1,909,000 |
| Amazon | $410,000 |
| Microsoft | $1,075,000 |
| Apple | $2,384,000 |

The skills you are building will be in high demand from companies, non-profits, government agencies, educational institutions, and more.

- The choice of how to spend your career is yours.

# Your Future Career

Quite a lot of you will likely end up working at some sort of technology company at some point in your life.

There's nothing (IMO) wrong with working at profit driven tech companies.

- Please do realize that even as a rank and file employee, you have the power to effect change, particularly if you are paid in stock (because then you are a partial owner).
  - Example: Sectarian violence in Myanmar driven by Facebook posts was a huge problem due to a lack of local language speakers. Nobody stepped up to fix this.
  - Example: Frances Haugen (former Facebook data scientist) leaked documents showing shady behavior by the company.

And of course realize that your skills can be used in many ways!

# Your Future Career

My request: Use your superpowers in a way that is a net positive for the lives of your fellow humans.

- I'm not saying that working for big companies is evil or anything, or that you have a moral obligation to work for relatively low wages for the public good (though that'd be great, of course).
- …. but keep in mind that your work will profoundly affect the world.

(Wanna talk about this stuff more? Take CS195!)

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CS61B Version 4.0 (Fall 2025)

- Many more smaller assignments, especially at the beginning.
  - No more project 0.
  - Not sure if this was successful. Maybe too tedious?
- Flexible points policy for HWs.
  - Previously we had a flexible deadlines policy for HWs.
- Gave vague permission for students to use LLMs on HWs.
- Midterm 1 as a checkpoint for knowing how to write code from scratch.
- Opt-in Attendance (ha).
- Focused course material even further on what matters (less syntax, less inheritance, less Java trivia).
- Renamed all the labs to just be HWs.

# Potential Changes for Sp26

Main changes I have in mind:

- Not teaching CS70 at the same time.
- Get students back in to rooms with each other (discussion attendance was 10% this semester, lecture was similar).
  - Target: 50% discussion attendance (which is what 61A is doing)
- Tentatively: More open-ended programming challenges for students with especially strong background.
- Tentatively: Pare back the swarm of early assignments.
  - Shift slightly back towards “flashy” assignments rather than “directly related to lecture” assignments.
- Tentatively: Midterm 1 is on a computer and retakable.

# Feedback Request

We read all of your feedback, especially on the end of course evaluations.

- There is an official form from the university, and an internal survey for 61B that we'll send out ourselves.

# AMA

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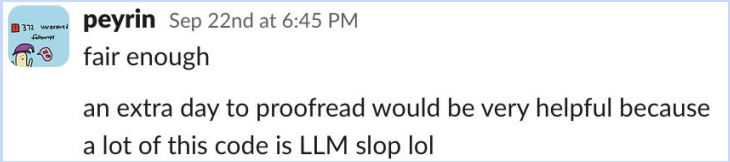
# Questions About Anything* for Josh and Peyrin?

What happened to Gitlet and BearMaps (old design projects)?
Do you use LLMs to create assignments?




How do I learn how to build cool things?
Can you discuss your cybercrimes?
How do you feel about the Neetcode 150 roadmap?
The software engineering market is volatile. How disposable are we?
Differences between engineering/industry job and academia job?
Has the market ever been this volatile since you started teaching?
Are post-GPT classes dumber than pre-GPT classes?
What is the successor to 61B? 61C feels more complementary than a sequel.
Favorite data structure?
What do you love about teaching?
Your most famous alumni?
Do people still create new data structures?
What is your best public speaking tip?
What's a class you haven't taught, but want to?
How do you keep up-to-date with learning new things about CS? (https://www.quantamagazine.org/)
If you teach 61B forever, will you teach other classes too?
For someone with a passion in activism, how do you not let your ideals go?

# So Long!

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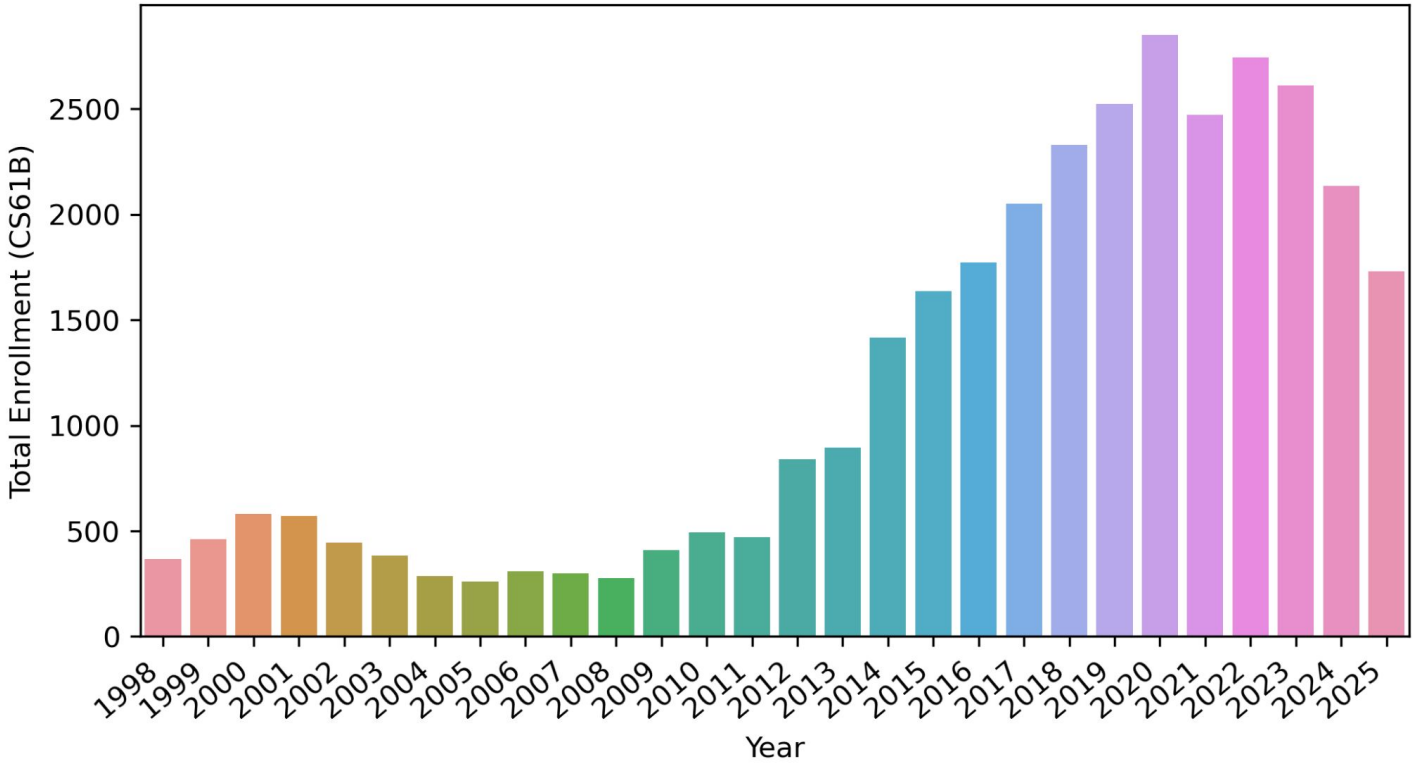
# 61B Would Love to Have You Next Spring

- Teaching interns: Learn more, help others, get units.
- Watch out for an announcement on the Fall 2025 61B Ed.



# Special Thanks to the Staff (who keep the wheels on the bus)`

**Head TAs**: Daniel Wang, Elaine Shu, Stacey Lei

TAs: Aditya Hariharan, Amy Wang, Anirudh Kotamraju, Dawn Schumacher, Iris Zhou, Karen Meng, Natalia Grondin, Ricardo Sandoval, Ritika Joshi, Ronnie Beggs, Ryan Yang, Tiffany Li, Yinqi Yi

Tutors: Andrew Schoenen, Avik Gautam, Ayush Gupta, Codey Ma, Dennis Koh, Isabel Dong, Jonathan Gasser-Brennan, Lawrence Wu, Lilah Durney, Marcus Koh, Michelle Koga, Miller Liu, Phoenix Wilson, Rushil Saraf, Sarveshan Saravanan

Teaching Interns (TIs): Alex Nava, Anay Shukla, Brendan Reyes, Deven Mittal, Elaine Liu, Emily Huynh, Ethan Yang, Ishanth Hombaiah, John Hu, Jonathan Cao, Krishna Chunduri, Kuljit Uppal, Monishka Tanwani, Saksham Jain, Sruthi Vegunta, Theo Sohn, Zhen Liu

# Closing Thoughts (Josh)

It is a terrifying and awesome responsibility to decide how you should spend hundreds of your precious hours here at Berkeley.

I hope I've done a good job. I know it wasn't perfect (and sadly never will be).

- (but I've got Ideas™)

… and thanks for all the kind words, feedback, and understanding when things are not as good as they could be.